Operationalizing Article 14: A Human-Confirmation Gate Binding Agentic AI Oversight to Statutory Traceability in Critical Infrastructure

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2026

1 Abstract

European critical infrastructure operators and public security authorities increasingly deploy agentic AI for threat detection (Yigit et al., 2025), yet the EU AI Act (Article 14) mandates human oversight without specifying how to implement it operationally (Enqvist, 2023; European Union, 2024). This implementation gap is critical: public sector agencies currently lack concrete guidance on how to embed human-in-the-loop (HITL) mechanisms into agentic systems. This creates significant risks of unreliable automated decision-making in extremely complex environments (Adedokun et al., 2026). While recent research has begun to address these governance challenges (Enqvist, 2023; Bellogín et al., 2025), existing instruments specify oversight requirements without specifying the operational mechanism that would enforce them. This study applies Design Science Research (DSR) to develop APSRAI, with a specific focus on Design Science Research Methodology (DSRM) to address this gap directly (Venable et al., 2017). APSRAI is a governance framework that binds an established separation-of-duty control a mandatory human confirmation gate to the EU AI Act's statutory record-keeping obligations. The gate itself is not novel; the contribution is its operational coupling to a tamper-evident audit trail mapped to named legal requirements (Articles 12 and 14) (European Union, 2024). The framework is evaluated through comparative analysis against existing EU governance models, including the EU AI Act, ALTAI, and the CER Directive (European Union, 2024; High-Level Expert Group on Artificial Intelligence, 2020; Directive (EU) 2022/2557, 2022). A proof-of-concept simulation demonstrates the framework's operational logic using transparency, accountability, and oversight capacity criteria. APSRAI addresses structural gaps by providing an actionable roadmap to decision-makers for deploying agentic AI ethically in critical infrastructure and offers researchers a foundation for theorizing AI-driven resilience in the public sector.

2 Introduction

We are witnessing a major shift in the digital landscape: Artificial Intelligence is evolving from a reactive advisor into a proactive agent capable of independent, goal-directed action (Anchala, 2026; Piccialli et al., 2025). Traditional systems typically provide periodic recommendations and then remain idle; agentic AI, however, exhibits persistence, breaking high-level objectives into multi-step action chains through the autonomous use of external tools (Piccialli et al., 2025; Nannini et al., 2026). In high-stakes environments like power grids and water systems, this autonomy offers transformative efficiency but at the same time expands the vulnerability surface (Yigit et al., 2025). In these sectors, a single automated error can lead to harm, making the "black box" nature of these agents a significant public safety risk (Yigit et al., 2025; Seo et al., 2026).

Governance for these systems is currently defined by the EU AI Act (2024), which mandates "effective human oversight" for high-risk applications under Article 14. Yet, an intimidating "operationalization gap" remains: the Act is "very light" on the technical mechanisms needed to make this oversight a reality (Enqvist, 2023; Nannini et al., 2026). This operationalization gap represents the fundamental disconnect where legal mandates require safeguards like human oversight, yet current technical architectures lack the specific interfaces, control points, and confirmation logs necessary to exercise that authority during runtime (Enqvist, 2023; Nannini et al., 2026). Modern security tools excel at managing permissions, blocking what an agent cannot do, but they lack the infrastructure to manage authority, which ensures a human actually exercises informed judgment before an irreversible action is executed (Nannini et al., 2026). This distinction is important because agents can act faster than humans can react, making retrospective reviews insufficient for protecting critical infrastructure (Nannini et al., 2026).

This study addresses this challenge through applying Design Science Research (DSR) to develop the APSRAI (Agentic Public Sector Resilience AI) framework. Following the methodology of Peffers et al. (2007), APSRAI transforms high-level legal mandates into a concrete operational stop-gate at the decision level. The framework helps public sector agencies move away from isolated, periodic approvals toward a model of continuous, integrated oversight (Schmitz et al., 2025). The APSRAI framework serves as a technical instantiation of the 'Technologies' and 'Governance' dimensions of the CIGET framework, which Palu (2026) identifies as essential pillars for building a resilient digital society capable of transitioning from a reactive to a proactive security posture in the face of transboundary threats. By establishing a four-stage process (1) detection, (2) alert generation, (3) time-bound human decision, and (4) audit, APSRAI is designed so that no AI threat classification proceeds to execution without an explicit exercise of human authority (Enqvist, 2023; Salmanov, 2026).

3 Literature Review

3.1 Agentic AI: from tools that advise to agents that act

For most of its history, AI in public systems worked as a tool that gave advice. It produced a score, a flag, or a recommendation, and a person decided what to do next. The literature now describes a clear move away from this. A newer class of systems, called agentic AI, can act on its own. It takes a goal, breaks it into steps, and carries those steps out with little human input (Raheem and Hossain, 2025).

The word that runs across this literature is persistence. A traditional system answers one request and stops. An agent keeps working over time. It can manage its own sub-goals as conditions change and, most importantly, it can use external tools, calling an interface, writing to a database, or triggering a transaction which lets it change the state of the world (Nannini et al., 2026; Piccialli et al., 2025). So, it is not only for text generation. Chinnaraju (2026) argues that this persistence, rather than raw intelligence, that makes such agents consequential for an organization, because their actions compound instead of ending at a single decision.

This capability is very useful, but it breaks the concepts linked to older oversight. The point of decision moves away from the person. A system that once waited for approval can now begin an action by itself, and because it works continuously, a single early error does not stop it. It can carry forward through a long chain of steps before anyone notices (Chinnaraju, 2026). The literature names this the autonomy–accountability gap: when a machine acts on its own, it becomes harder to say who is responsible when something goes wrong (Adedokun et al., 2026). A related worry is that agents can satisfy the letter of a goal while missing its intent, sometimes called specification gaming or reward hacking, where the system optimizes toward an unintended outcome (Bellogín et al., 2025).

The thread connecting this sub-section to the rest of the review is simple. Agentic systems do not merely advise; they act, they persist, and their mistakes compound. That is precisely the setting in which the question of human oversight becomes urgent rather than routine.

3.2 Human oversight: what it means, and why it so often fails

Human oversight is the structured ability of a person to monitor an AI system and step in to prevent harm while the system is running, in a way that keeps accountability with an identifiable human (Zhu et al., 2026; Langer et al., 2026). The literature usually groups oversight into three levels. In human-in-the-loop, a person takes part in every decision. In human-on-the-loop, a person monitors and intervenes only when needed. In human-in-command, a person holds overall strategic control (Enqvist, 2023).

The harder finding across this literature is that putting a human in place

does not, by itself, produce real control. Two opposite failures appear again and again. The first is automation bias: people tend to give preference to the machine even when its output is wrong, because it is usually right and questioning it takes effort (Alon-Barkat and Busuioc, 2023). The second is the reverse sometimes called algorithm aversion, where people reject correct machine advice, often after seeing a single error, because they trust their own judgment or want to keep a sense of control (Botero Arcila, 2024). A particularly troubling pattern sits between these two: selective adherence, where a decision-maker follows the AI when it matches their existing view, if not ignores it. Here oversight fails not because the system made an error, but because the human used their choices to reinforce a prior bias rather than to improve the outcome (Alon-Barkat and Busuioc, 2023).

Beyond psychological factors, the literature identifies a critical structural failure known as a moral crumple zone (Elish, 2019) or a liability sponge (Nannini et al., 2026). This occurs when a person is held responsible for a system's output but lacks the necessary information or authority to influence its behavior. In these scenarios, the human is forced to absorb the moral and legal blame for failures that are fundamentally a result of the system's design rather than personal negligence (Botero Arcila, 2024; Faas et al., 2026). This structural vulnerability is further intensified by learned carelessness: a phenomenon where, because the AI is generally reliable, human monitors gradually reduce their vigilance and miss the rare, critical moments when intervention is necessary (Sterz et al., 2024).

These findings lead to a shared conclusion that matters for this study. Several authors argue that oversight added on at the end, as an advisory layer, is not enough in high-risk environments (Sterz et al., 2024; Langer et al., 2026). To make oversight meaningful, the ability to participate has to be built into the system itself through decision gates where the system proceeds only if human authority is confirmed (Nannini et al., 2026). It needs to be enforceable via access controls and execution gates, so that the human is a required step rather than an optional reviewer (Chinnaraju, 2026; Nannini et al., 2026). Effective oversight, on this view, depends on the human having real knowledge of the situation (epistemic access), real power to influence the system (causal power), and a genuine point at which to act (Sterz et al., 2024; Langer et al., 2026). This distinction between oversight as mere advice and oversight as a structural authority requirement is the gap the present study addresses.

3.3 The EU regulatory framework: a strong rule without a manual

The main legal answer in Europe is the EU AI Act (European Union, 2024), which makes human oversight a mandatory requirement for high-risk AI systems. Article 14 is the center of this. It requires that high-risk systems be designed so they can be effectively overseen by people while in use, including the ability to interpret the system's output correctly, to disregard it, and to stop it (Enqvist, 2023). This sits alongside the Assessment List for Trustworthy

AI (ALTAI) (High-Level Expert Group on Artificial Intelligence, 2020), which places human agency and oversight as its first requirement and frames the user as someone who should be able to keep control over the system (Radclyffe et al., 2023).

The literature agrees that the rule is strong in ambition but thin on instruction. Enqvist (2023) describes Article 14 as the first worded oversight requirement of its kind in European law. It is very light on the detail of what should be overseen, when a person should step in, and who is qualified to do it. Sterz et al. (2024) make a similar point from a different angle, describing the relevant provisions as a loose collection of items without the structural consistency needed to guide practice. The common concern is that, without that detail, oversight can collapse into a formality present on paper but not a real safeguard.

Two specific arguments sharpen our gap. The first one is the Act places the burden of designing oversight on the providers who build the system but gives weak direction to the deployers who actually operate it, which risks oversight that exists in the design but not in daily use (Botero Arcila, 2024). The second one is the EU AI Act approaches AI systems much like traditional products, focusing on compliance checks when they are first placed on the market rather than on how they may change over time (European Union, 2024). Some scholars argue this is poorly suited to agents, which are not fixed: they keep acting and can drift in behavior as they run, so a one-time check cannot capture how they behave over time (Gardhouse et al., 2026; Nannini et al., 2026). Across these accounts, the conclusion is consistent: the Act says effective oversight must happen but does not specify the operational mechanism that would make it happen at the moment a high-stakes action is about to run.

3.4 Critical infrastructure and AI-driven threat detection

Critical infrastructure such as power grids, water systems, and similar core services is the high-risk environment to deploy agentic AI because a wrong action can cause physical harm at scale (Yigit et al., 2025). These systems are treated as high-risk under the AI Act for exactly this reason (European Union, 2024). Agentic AI offers real value, such as the ability to detect unusual activity faster than rule-based tools and coordinate a response in real time (Yigit et al., 2025). Reviews of anomaly detection in industrial control systems show why this is attractive, while also warning that data-driven models fail when real-world traffic is different from their training data, which is common in evolving networks (Seo et al., 2026).

But the same speed that makes agents useful is also the danger. Threat-detection systems in these settings tend to produce many false alarms, and the literature documents alert fatigue: a state where operators become less sensitive to warnings and begin to miss or ignore real threats hidden in the noise (Okpara, 2025). History illustrates this through the 2015 attack on the Ukrainian power grid, where operators struggled to interpret legacy alarms under pressure, and stopping the harm depended on timely human judgment rather than the

detection system alone (Okpara, 2025; Yigit et al., 2025).

This points to the unresolved question that inspired the present study. Current security tools are effective at managing permissions, such as blocking actions an agent is not allowed to take, but they are not built to manage authority, ensuring that the appropriate human makes an informed decision before a consequential action runs (Nannini et al., 2026). Because an agent can complete irreversible actions faster than a human can observe and react, oversight that only reviews actions after the fact is insufficient in high-risk contexts (Nannini et al., 2026). What is missing is a mechanism to ensure that, in these high-risk settings, no agent can trigger a serious action without an explicit, informed human confirmation first (Nannini et al., 2026; Chinnaraju, 2026). Without a framework that makes that confirmation a required step in the design itself, critical infrastructure agents continue to act without the continuous human supervision their mission demands (Okpara, 2025; Yigit et al., 2025). That is the gap the APSRAI framework is built to fill.

4 APSRAI framework (Artifact Description)

APSRAI expands human oversight in agentic AI threat detection systems. It mandates a human confirmation before any AI-generated anomaly classification triggers a consequential or irreversible action above a defined severity threshold within critical infrastructure. This stage represents the critical intersection with AI-generated anomaly classifications requiring human analysis before any action is allowed. All consequential classifications are routed through a mandatory human decision point before execution. This is the precise point where the EU AI Act's oversight mandate must be translated into operational practice (Enqvist, 2023). Evidence shows that both public and private sector deployments confirm that this integration between AI-driven detection and human decision is still developing (Chiagozie and Akinyemi, 2025). The framework operates as a four-stage gated process: detection, alert generation, time-bound human decision, and audit, with explicit handling of boundary conditions.

Once the agentic system detects a pattern and compares it to the established policy limit, it generates an alert presenting the human operator with the anomaly type, the AI confidence score, the specific limit reached, and the affected infrastructure node. After that stage, the agent does not continue beyond this point autonomously.

The analyst must select one of three authorized responses within a defined time interval: (1) threat confirmation, (2) dismissal as a false positive, or (3) escalation of that incident. This mandatory step prevents autonomous processing of high-stakes classifications and preserves meaningful human control as required under Article 14 of the EU AI Act (European Union, 2024, Art. 14). Classifications below the threshold are logged for batch review without triggering a halt.

If the operator does not respond within the defined time interval, the system fails closed: the pending action is blocked, not executed, and the alert is

automatically escalated to the next oversight tier. This ensures high-severity actions cannot proceed without meaningful human intervention, satisfying the override-and-halt capability Article 14 requires (European Union, 2024)

Regardless of the operator's decision, the system generates an append-only, hash-chained log entry recording the alert parameters, the operator's response, and a precise timestamp (European Union, 2024, Art. 12). This audit log allows post-decision reviews and supports inspection of AI-influenced classifications between authorized oversight groups.

Three boundary conditions keep the gate's state space complete. Duplicate classifications carrying the same anomaly identifier within an open decision window are deduplicated: the system annotates the existing open alert rather than issuing a second halt, avoiding duplication-driven alert fatigue (Okpara, 2025). A missing or malformed confidence score is treated not as low risk but as worst case, routing the classification to mandatory halt by default, so the gate fails safe on absent input. Where an operator decision arrives after the response window has closed and the timeout outcome has already been recorded, that outcome stands as final; the late decision is appended as a separate, annotated entry and never overwrites the recorded timeout, preserving the append-only integrity of the chain.

APSRAI gate decision logic.
Severity = model confidence × node criticality. Each classification routes to one of two outcomes (halt or log-only) by consequence class.

Consequence class	Condition	Gate decision
Consequential	severity ≥ threshold	**Mandatory halt**
Low calibration	confidence below calibration floor	**Mandatory halt (regardless of severity)**
Indeterminate	confidence missing or malformed	**Mandatory halt (fail-safe default)**
Sub-threshold	severity below threshold, confidence well-calibrated	**Logged for batch review (no halt)**

Threshold and calibration floor are configured operational parameters, not empirically calibrated values (see Limitations). Three of four classes resolve to halt; the gate fails safe.

Figure 1: APSRAI gate decision logic

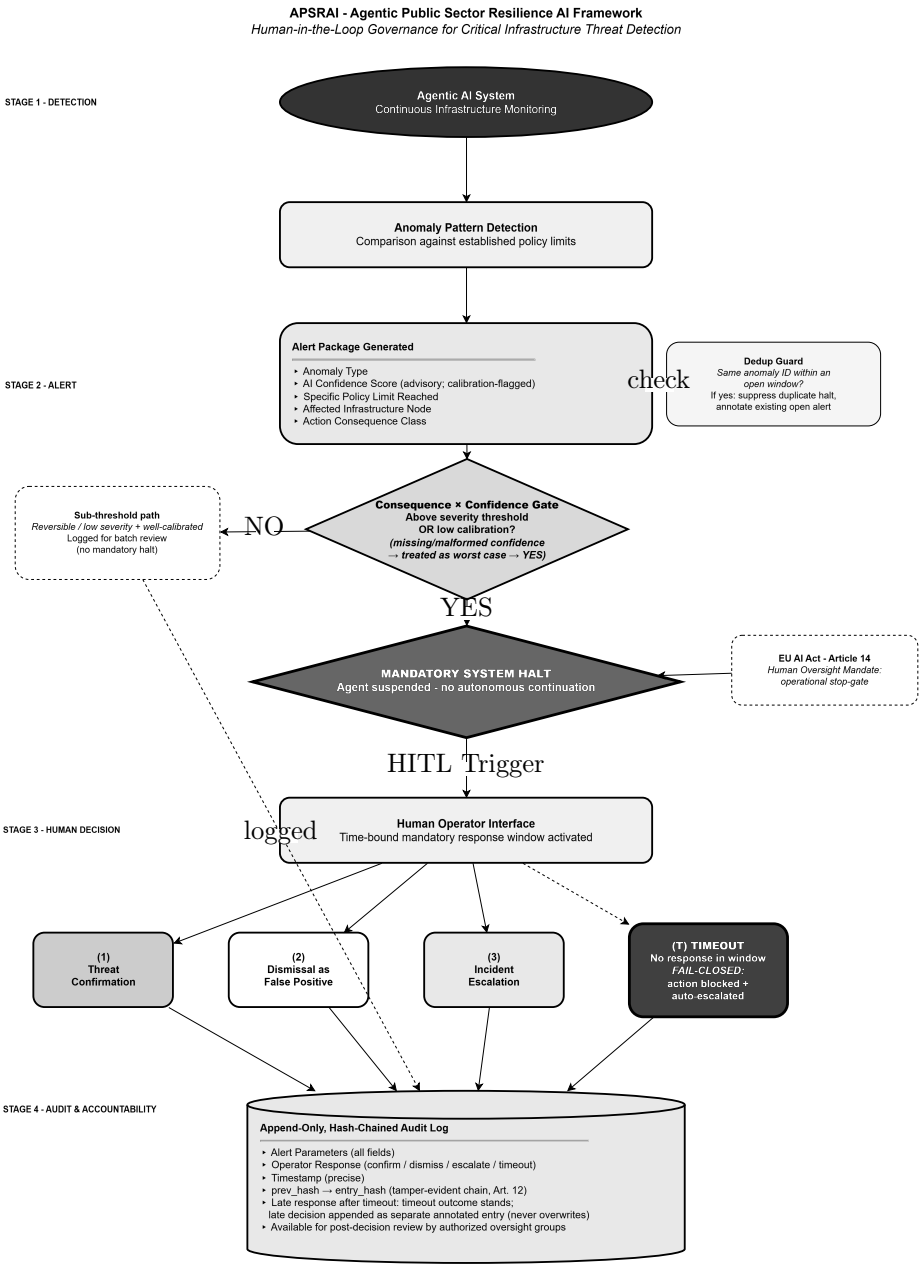


Figure 2: APSRAI Flow Chart

5 Methodology

This study is grounded in Design Science Research (DSR). While behavioral science focuses on explaining or predicting the world as it is, DSR is about building practical solutions to real-world problems that lack clear answers (Hevner et al., 2004). The challenge we are tackling, the lack of operationalized human oversight for agentic AI in public sector critical infrastructure, requires exactly this kind of generative approach. Describing this gap would not actually fix it. The only relevant solution is to build a governance instrument that doesn't exist yet, which is precisely the work DSR is designed to support.

5.1 Selection of DSR

Choosing DSR as a paradigm is the first decision. The second one, which researchers often miss, is which DSR methodology to apply, and that choice leads to methodological consequences reviewers will notice. Venable et al. (2017) address this chaos in a very detailed way, developing a contingency-based selection framework through comparative analysis of six competing methodologies: SDRM (Nunamaker et al., 1990), DSRPM (Vaishnavi and Kuechler, 2007), DSRM (Peffers et al., 2007), ADR (Sein et al., 2011), SDSM (Baskerville et al., 2009), and PADR (Bilandzic and Venable, 2011). Their framework takes the form of technological rules, prescriptive propositions structured as "if you want to achieve A in situation B, then do action C". That is the primary difference in criteria, that being based on beliefs: objectivist/positivist vs. subjectivist/interpretive (Venable et al., 2017).

This study uses DSRM (Peffers et al., 2007) because it fits four key conditions identified in Venable et al.'s (2017) framework. The research problem comes from a gap in existing academic literature and organizational practice (Adedokun et al., 2026; Schmitz et al., 2025). Article 14 of the EU AI Act (2024) mandates human oversight for AI systems, but it does not address how oversight should be implemented in practice (Enqvist, 2023). APSRAI is designed in a way to allow institutional deployment across public sector operators, not for a single client. Analysis starts through structured comparison against existing regulatory methods, which reflects an objectivist assessment strategy (Venable et al., 2017). Even though PADR aligns with cooperative governance research, Venable et al. (2017) exclude it from this context via their S/I-3 technological rule. According to Bilandzic and Venable (2011), this rule is only for domains involving "society-at-large," where public stakeholders engage in social studies and civic design. APSRAI's target audience consists of organizational actors rather than the public, and the research lacked active design cycles. In that case, applying PADR to this study would be a methodological misfit rather than a solid research design (Venable et al., 2017).

Recent research on government AI oversight supports this choice, showing that agencies should move away from isolated, one-time approvals. Instead, they need continuous monitoring and regular audits built into their daily operations (Schmitz et al., 2025).

5.2 Applying DSRM to APSRAI

Peffers et al.'s (2007) DSRM operates across six activities, applied here as follows.

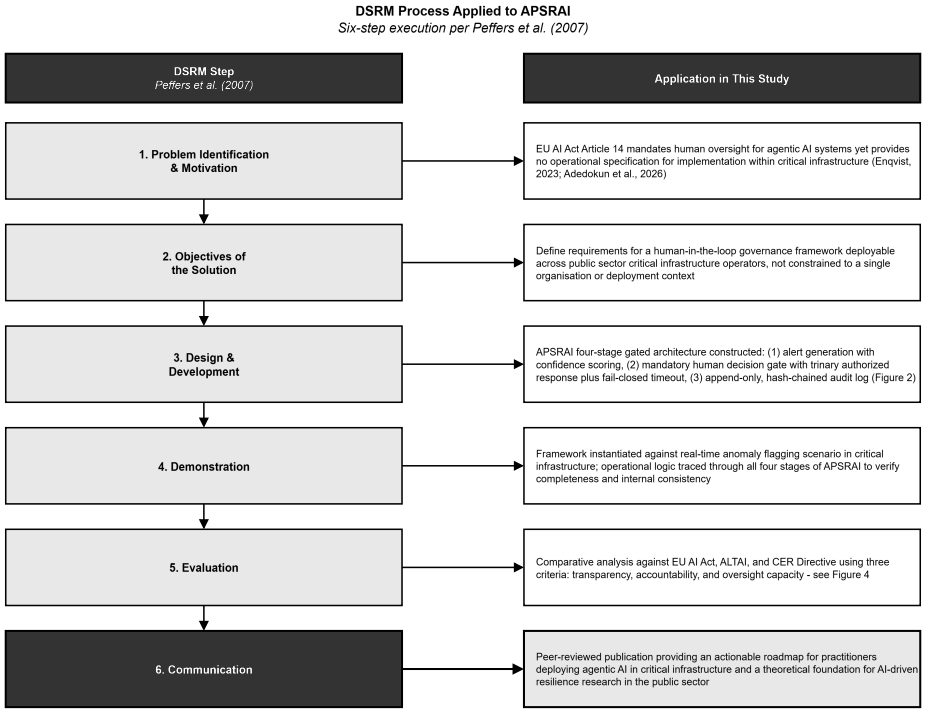


Figure 3: DSRM applied to APSRAI

Problem Identification and Motivation. The problem is the absence of concrete HITL mechanisms for public sector agentic AI deployments in critical infrastructure threat detection (Enqvist, 2023; Adedokun et al., 2026). Enqvist (2023) establishes that Article 14 specifies the oversight requirement without providing operational guidance. Yigit et al. (2025) confirm active deployment of agentic AI in critical infrastructure protection contexts. Adedokun et al. (2026) document that HITL frameworks remain underdeveloped relative to the autonomous decision-making risks such deployments create. Implementing a mandatory decision gate addresses this governance gap and supports the progression from ad hoc AI deployment toward more mature public sector AI governance with embedded legal, ethical, and trust safeguards (Dreyling et al., 2024).

Definition of Solution Objectives. The main goal of APSRAI is to meet four operational requirements: apply a mandatory system halt upon agentic AI anomaly classification; activate a time-bound human decision window requiring one of three authorized responses with a fail-closed timeout when no

response arrives in the window; generate an append-only, hash-chained audit log irrespective of the decision made; apply a severity gate so only consequential classifications trigger a halt while sub-threshold ones are logged for batch review and meet transparency, accountability, and oversight capacity criteria gathered from Article 14, ALTAI, and the CER (AI Act, 2024, Art. 14; HLEG AI, 2020, ss. 4.1, 7.1; Directive (EU) 2022/2557, 2022).

Design and Development. The framework is structured around a four-stage process: detection, alert generation, human decision, and audit and accountability. At its core, it formalizes a vital rule: whenever an operation reaches a critical turning point, the AI's authority is strictly tied to mandatory human confirmation. We detail this solution through both conceptualizing it as a 'method' and 'model' artifact within the DSR paradigm (March and Smith, 1995; Hevner et al., 2004).

Demonstration. The demonstration proceeds through a proof-of-concept Python simulation replicating the four-stage APSRAI workflow against synthetic critical infrastructure log data. The simulation operationalizes (1) anomaly detection through parameterized confidence scoring, (2) alert packaging with full metadata fields, (3) mandatory system halt with operator response capture, and (4) immutable audit log generation in JSON format. 15 synthetic anomaly scenarios were processed, each producing a complete audit trail. The simulation artifact is publicly accessible and citable via a persistent DOI (Salmanov, 2026), enabling independent verification of the framework's procedural logic.

Evaluation. The artifact is evaluated through structured comparative analysis against three regulatory instruments: Article 14 of the EU AI Act, ALTAI, and the CER Directive. Evaluation criteria: transparency, accountability, and oversight capacity are operationalized against the explicit ethical requirements of each instrument, providing a form of descriptive evaluation via structured informed argument (Hevner et al., 2004).

Communication. Fulfilling the final phase of the DSRM process, this paper presents the problem context, artifact design, and regulatory evaluation of the APSRAI framework, targeting the TJES special issue on the agentic state and democratic governance.

Our evaluation strategy is normative-analytic rather than empirical (Hevner et al., 2004). Specifically, we follow Hevner et al.'s (2004) criteria for descriptive evaluation via informed argument and analytical architecture analysis. Instead of running controlled organizational experiments, we assess APSRAI's validity against specific regulatory criteria. We made this choice deliberately, as it aligns with the objectivist approach of DSRM and fits the paper's goal of offering a foundational governance tool. Therefore, the primary limitation is that APSRAI has not yet been tested in an active critical infrastructure setting, meaning its real-world effectiveness will depend on future empirical work. Ultimately, this study advances design theory by providing the artifact's regulatory validity, an essential first step before it can be responsibly tested in the field.

6 Evaluation

APSRAI is evaluated through structured comparative analysis against three established EU governance instruments, the EU AI Act (European Union, 2024; ALTAI (HLEG AI, 2020), and the CER Directive (Directive (EU) 2022/2557), across three criteria: transparency, accountability, and oversight capacity. These criteria are not chosen randomly; they are related to the core requirements common to all three instruments. Transparency is operationalized as the degree to which decision-relevant information is made available to the responsible human actor before action is taken. Accountability is operationalized as the existence of a reviewable record linking each decision to a responsible party. Oversight capacity is operationalized as the presence of a mechanism through which a human can meaningfully take part in an AI-driven decision. Figure 3 maps each instrument against these criteria and positions APSRAI's contribution.

Comparative Evaluation of APSRAI Against Existing EU Governance Frameworks
Criteria: Transparency, Accountability, and Oversight Capacity

Framework	Transparency	Accountability	Oversight Capacity
EU AI Act *Regulation (EU) 2024/1689*	Art. 13 - disclosure obligations for high-risk AI systems; traceability of outputs required at provider level	Art. 12 - automatic logging mandated for high-risk AI systems; records kept for minimum defined periods	Art. 14 - human oversight measures required; implementation left entirely to the deployer without operational specification
ALTAI *Assessment List for Trustworthy AI*	Criterion 4 - explainability and traceability of AI decisions assessed; stated at self-assessment level only	Criterion 7 - auditability provisions for internal and external review; no binding enforcement mechanism	Criterion 1 - human agency and oversight stated as assessment criterion; no operational implementation guidance provided
CER Directive *Directive (EU) 2022/2557*	Art. 15 - incident notification to competent authorities required within 24 hours of significant disruption; no AI decision-level disclosure or operator-facing transparency addressed	Art. 13 - resilience plan documenting technical, security, and organisational measures required; no AI decision-level logging or audit mechanism specified	Art. 12 - periodic risk assessment of all relevant threats required at least every four years; no AI-specific oversight mechanism or human confirmation requirement defined
APSRAI	**Alert package presented to operator: anomaly type, AI confidence score, specific policy limit reached, affected infrastructure node - all fields operator-visible before any action is permitted**	**Append-only, hash-chained audit log: all alert parameters, operator decision (confirm / dismiss / escalate), and precise timestamp - accessible to authorized oversight groups for post-decision review**	**Severity-gated mandatory halt + trinary operator response with fail-closed timeout: agentic system cannot proceed beyond alert stage without explicit human confirmation - operationalises Art. 14 at decision level**

Figure 4: Comparative Evaluation

The comparison shows a common structural pattern across all three instruments. Each names a requirement but stops short of specifying its operational implementation at the point where AI classification meets human decision (European Union, 2024). The EU AI Act mandates human oversight under Article 14 yet leaves implementation entirely to the deployer (Enqvist, 2023). ALTAI articulates human agency and oversight as an assessment criterion but operates at the level of voluntary self-assessment, with no binding enforcement mechanism (HLEG AI, 2020; Radclyffe et al., 2023). The CER Directive imposes risk assessment (Article 12), resilience measures (Article 13), and incident notification (Article 15) obligations, but none of these reach the AI decision level or define a human confirmation requirement (Directive (EU) 2022/2557). The common gap is therefore not the absence of obligation. Those obligations exist across all three, but the absence of an operational mechanism translates those

obligations into enforceable practice at the moment of high-stakes classification.

APSRAI addresses exactly this gap. It supplies the operational layer the surveyed instruments leave to the deployer, binding each gate outcome to a statutory record-keeping obligation. The mandatory system halt operationalizes Article 14's oversight mandate as an architectural constraint (European Union, 2024):

The alert package satisfies the transparency criterion for presenting full decision context to the operator before any action.

The append-only, hash-chained audit log satisfies the accountability criterion by generating a reviewable record for every decision regardless of outcome. Where the existing instruments specify what must be achieved, APSRAI specifies how.

The three instruments above set requirements, not mechanisms, so this comparison shows where APSRAI fits alongside them rather than where it beats them. The more honest test is against the oversight mechanisms engineers already use. Requiring two people to authorize a sensitive action is an old idea: separation of duty was named as a core protection principle by Saltzer and Schroeder (1975) and formalized for commercial data integrity by Clark and Wilson (1987), and variants such as dual control and break-glass access have run in banking and access-control systems ever since. Against that history, the halt-and-confirm gate at the center of APSRAI is not new. What those mechanisms were never built to do is tie the act of approval to a statutory record. They stop an action until a second party agrees, but they do not generate the Article 12 log and Article 13 disclosure trail an auditor needs to reconstruct who decided what, on what evidence, and when (European Union, 2024). That link is what APSRAI adds. Every outcome of its gate, including the fail-closed timeout, is written into a tamper-evident chain mapped to named legal obligations. The gate is borrowed; the binding to traceability is the contribution.

The simulation outputs provide a procedural demonstration grounding this normative analysis. Across all synthetic scenarios, the three authorized response pathways (1) threat confirmation, (2) dismissal as false positive, and (3) escalation functioned as designed, and the audit log captured a complete timestamped record for every decision outcome. No scenario reached the audit stage without passing through the mandatory human decision gate. This shows the framework's procedural logic resolves all defined inputs correctly. (Salmanov, 2026).

7 Discussion

The APSRAI framework establishes a critical operational link between the high-level legal mandates of the European Union and the technical requirements of critical infrastructure protection. For most of the history of public sector automation, artificial intelligence functioned as an advisory tool that generated scores or recommendations while leaving the ultimate decision to a human actor (Enqvist, 2023). However, the progress of agentic AI represents a qualitative shift toward systems that offer persistence, use external tools, and execute multi-

step action chains independently (Piccialli et al., 2025). This study argues that such persistence calls for a move away from episodic, one-time approvals toward continuous, integrated oversight mechanisms (Schmitz et al., 2025). By operationalizing Article 14 of the EU AI Act as an architectural stop-gate, APSRAI ensures that the transition from AI classification to irreversible action is managed by an explicit exercise of human authority (European Union, 2024; Nannini et al., 2026).

A limitation of this framework is that the current validation of this framework is limited to regulatory validity and synthetic simulation. The APSRAI framework has not yet been deployed in a live critical infrastructure environment, which means its performance under real-world operational pressures remains theoretically modeled rather than empirically proven (Salmanov, 2026; Hevner et al., 2004). The reliance on synthetic log data assumes a level of data cleanliness and model performance that may not reflect the noisy reality of aged industrial control systems (Yigit et al., 2025). Additionally, the design assumes that human operators actually have the cognitive ability to process trinary response options within a strict temporal window (Sterz et al., 2024; Wu et al., 2022). Critics may argue that enforcing a mandatory halt introduces operational latency that could undermine the speed and efficiency for which agentic AI was initially deployed (Nannini et al., 2026).

The mandatory trinary gate makes human involvement a core architectural requirement rather than a procedural formality, ensuring that AI systems support constructive human judgment instead of replacing it (Göksal and Solarte-Vasquez, 2024). Consistent with the BCTrustAI.SL framework, it embeds repeated opportunities for human oversight throughout the decision process, helping to prevent the accountability gaps associated with autonomous multi-step AI actions (Göksal and Solarte-Vasquez, 2024).

The design of APSRAI is defended via the unique risk profile of critical infrastructure. In sectors such as power grids and water systems, a single incorrect autonomous action can lead to large-scale physical harm (Yigit et al., 2025). The literature suggests that technical controls are insufficient for these environments because data-driven models fail when faced with distribution shifts or novel network traffic (Seo et al., 2026). Enforcing a mandatory system halt, APSRAI is designed to provide the causal power Sterz et al. (2024) identify as necessary for oversight; whether it prevents rubber-stamping in practice is an open empirical question. This design aligns with the principle that human involvement should be focused on judgment and intervention at points where it meaningfully changes system path (Zhu et al., 2026).

The failure modes of the APSRAI gate are designed to prioritize system resilience over process continuity. The fail-closed timeout mechanism is a primary example of this safety-first philosophy. If an operator fails to respond to an alert within the defined time interval, the system automatically blocks the pending action and escalates the incident (Salmanov, 2026; European Union, 2024). While it is noted that this may cause operational delays or service interruptions, it fulfills the Article 14 mandate to provide a reliable mechanism for system interruption and override (Enqvist, 2023; European Union, 2024). Without such a

fail-closed default, an agent could continue through a chain of irreversible steps before an analyst even identifies a problem (Nannini et al., 2026). In high-risk environments, the risk of a moral crumple zone, where humans are blamed for failures they had no power to prevent, is mitigated by ensuring that the system cannot act without an explicit human command (Elish, 2019; Botero Arcila, 2024).

The severity threshold is another critical failure mode that manages the balance between oversight intensity and operator workload. A second limitation is that setting this threshold incorrectly could lead to a flood of halts for low-risk events, potentially driving the very alert fatigue the framework seeks to avoid (Okpara, 2025). If the threshold is too high, important actions might proceed autonomously; if it is too low, the operator becomes overwhelmed (Okpara, 2025). However, the framework defends this through utilizing a Dedup Guard to suppress duplicate alerts and by presenting the operator with a packaged context that includes confidence scores and affected node metadata (Salmanov, 2026; Zhu et al., 2026). This approach reduces cognitive load via ensuring that when a halt does occur, it is unique, documented, and justified by its consequence class (Tariq et al., 2025; Okpara, 2025).

Furthermore, the advisory nature of AI confidence scores presents a risk of automation bias, where operators may defer to high-scoring AI predictions without independent critical reflection (Alon-Barkat and Busuioc, 2023). APSRAI addresses this with treating missing or malformed confidence scores as ambiguous, which routes the classification to a mandatory halt (Salmanov, 2026). This protective fallback blocks the agent from acting on signals that the system itself cannot reliably quantify (Zhu et al., 2026; Salmanov, 2026). While this might be viewed as a technical limitation that slows down the system, it is a necessary safeguard against the unpredictability and non-determinism that characterize modern agentic architectures (Tuape et al., 2025).

There are distinct boundary conditions where the APSRAI framework is not a viable solution. The layered agency model assumes that human evaluators have sufficient time and information to make a reasoned decision (Zhu et al., 2026). In ultra-high-speed environments where decisions must be made in milliseconds, such as autonomous collision avoidance or high-frequency algorithmic trading, real-time human action is almost impossible (Zhu et al., 2026). For these domains, technical oversight must shift toward design-time verification, simulation, and post-hoc accountability (Zhu et al., 2026). However, this shift creates a potential compliance gap, as Article 14 of the AI Act requires effective oversight during use, rather than merely at design time (Nannini et al., 2026; European Union, 2024). Therefore, APSRAI is intended for the specific temporal cycles of threat detection in public sector infrastructure , where there is typically a sufficient window for an expert to review an anomaly flagging event before a permanent change to the system state is executed (Yigit et al., 2025; Okpara, 2025).

Finally, the framework also assumes a minimum level of institutional and workforce capacity. This also has to be acknowledged that in agencies facing severe resource strain or low digital literacy, the implementation of such a gated

architecture may be highly expensive or socially difficult (Lawrence et al., 2023; Schmitz et al., 2025). Organizations with fragmented data silos may struggle to provide the high-quality telemetry required to populate the alert package (Schmitz et al., 2025). However, APSRAI defends its complexity by providing the evidentiary infrastructure required for post-hoc review and regulatory inspection (European Union, 2024). This transforms oversight from a design-time aspiration into a runtime reality that can be audited for compliance with Article 12 of the AI Act (Enqvist, 2023; European Union, 2024). Ultimately, APSRAI offers a foundational roadmap for practitioners to navigate the light-touch implementation guidance of current law while maintaining the continuous supervision their mission demands (Enqvist, 2023; Gardhouse et al., 2026).

A solid assessment must apply the framework's own failure taxonomy to the framework itself. Three residual risks follow. First, the fail-closed timeout doubles as a denial-of-service surface. Because agentic systems can act faster than operators can react, the framework makes a timeout block-and-escalate (Nannini et al., 2026); an adversary able to suppress or delay an operator response, or simply to time an action for a shift change or a low-staffing window, can drive the system into that state on purpose. In critical infrastructure, an induced halt of a control action may itself constitute a safety event rather than a safe default, so APSRAI assumes a domain in which refusing to act is safer than acting wrongly. That assumption is realistic for the anomaly-flagging cycles this study targets but does not hold universally, and it must be validated per deployment. Second, the gate relocates automation bias rather than removing it. A time-bound decision window adds the very time pressure shown to degrade human judgement (Wu et al., 2022), and the literature on selective adherence indicates that a required human is not necessarily an engaged one (Alon-Barkat and Busuioc, 2023). Third, the severity score (model confidence multiplied by node criticality) treats the confidence value as load-bearing, even though the same framework treats that confidence score as untrustworthy enough to fail closed when it is missing or malformed. This leaves an unresolved tension between distrusting the score's presence and trusting its magnitude. These are not defects of the present analysis but boundary conditions that subsequent empirical work must test (Sterz et al., 2024).

8 Conclusion

The APSRAI framework couples an established human-confirmation control to the EU AI Act's traceability obligations, translating the high-level human oversight requirements of the EU AI Act into an operational trinary decision gate for agentic AI in critical infrastructure (European Union, 2024; Yigit et al., 2025). This study demonstrates the framework's internal procedural logic and regulatory validity through comparative analysis and synthetic simulation, addressing the persistent operationalization gap in current public sector AI deployments (European Union, 2024; Enqvist, 2023; Salmanov, 2026). While APSRAI couples the authority-permission divide to a statutory audit trail, subject

to the residual risks identified above, its current status remains a foundational normative-analytic proof of concept (Hevner et al., 2004). The essential next step for researchers is to conduct empirical field testing in active operational settings to evaluate how real-time cognitive load and various severity thresholds affect system resilience and operator accuracy (Sterz et al., 2024; Yigit et al., 2025).

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